English to Konkani Machine Translation for the Health Domain

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Declaration

I hereby declare that the data presented in this Dissertation report entitled "English to Konkani Machine Translation for the Health Domain" is based on the results of investigations carried out by me in the Discipline of Computer Science and Technology at Goa Business School, Goa University, under the supervision of Prof. Jyoti D. Pawar, and the same has not been submitted elsewhere for the award of a degree or diploma by me. Further, I understand that Goa University or its authorities will not be responsible for the correctness of observations, experimental or other findings given in the dissertation. I hereby authorize the University authorities to upload this dissertation on the dissertation repository.

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Radhika Ramnath Dabholkar

Abstract

Machine translation for low-resource Indian languages in specialised domains remains an open and largely unaddressed problem. While significant progress has been made for major Indian languages such as Hindi and Bengali, languages like Konkani despite holding official scheduled status and being spoken by approximately 7.6 million people have received little attention in domain-specific natural language processing research. This gap is particularly consequential in the health domain, where medical instructions, dosage information, and clinical advice require precision that general-purpose translation systems often fail to provide.

This dissertation investigates whether meaningful English-to-Konkani health-domain machine translation can be achieved using currently available pretrained models and a modest amount of domain-specific parallel data. The central argument of this work is that general-purpose systems fail not because the task itself is infeasible, but because strong neural models have not been adequately adapted to this specific language domain combination.

To establish the current baseline, three existing systems Google Translate, Microsoft Bing Translator, and IndicTrans were evaluated on 992 health-domain sentence pairs. All three systems scored below BLEU 7.0, confirming that currently available tools are inadequate for this use case. A domain-adapted approach was then developed using the IndicTrans2 base model fine-tuned on approximately 19,794 parallel English–Konkani health-domain sentences from the ILCI corpus. This was combined with a preprocessing pipeline designed to protect medical entities, treat compound health terms as single translation units, and preserve verb structure through part-of-speech-based chunking.

The resulting fine-tuned system substantially outperformed all existing baselines, achieving a BLEU score of 21.85 and a chrF score of 54.67 on the held-out ILCI test set. These improvements are not marginal; they demonstrate that domain adaptation is not merely beneficial in this context, but necessary for meaningful translation quality.

The broader implication of this work is that low-resource need not mean unserved. With an appropriate pretrained foundation, thoughtful preprocessing, and domain-specific data, even a single GPU and a modest parallel corpus can produce significant gains in translation quality for a language community that currently lacks resources tailored to its healthcare needs.

Keywords: Machine Translation; Konkani; Low-Resource NLP; Health Domain; IndicTrans2; BLEU; chrF; TER; Domain Adaptation.

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Abbreviations

BLEU	**B**ilingual **E**valuation **U**nderstudy
BMI	**B**ody **M**ass **I**ndex
BPE	**B**yte **P**air **E**ncoding
chrF	**Ch**aracter n-gram **F**-score
CPU	**C**entral **P**rocessing **U**nit
DHA	**D**ocosahexaenoic **A**cid
EPA	**E**icosapentaenoic **A**cid
FP16	16-bit **F**loating **P**oint Precision
GPU	**G**raphics **P**rocessing **U**nit
HDL	**H**igh-**D**ensity **L**ipoprotein
ILCI	**I**ndian **L**anguages **C**orpora **I**nitiative
LDL	**L**ow-**D**ensity **L**ipoprotein
MT	**M**achine **T**ranslation
MWE	**M**ulti-**W**ord **E**xpression
NER	**N**amed **E**ntity **R**ecognition
NLP	**N**atural **L**anguage **P**rocessing
NMT	**N**eural **M**achine **T**ranslation
POS	**P**art **o**f **S**peech
TAM	**T**ense-**A**spect-**M**odality
TER	**T**ranslation **E**dit **R**ate
VRAM	**V**ideo **R**andom **A**ccess **M**emory
WMT	**W**orkshop on **M**achine **T**ranslation

Chapter 1

Introduction

1.1 Background

Healthcare depends on clear communication. When a patient cannot understand a diagnosis, a dosage instruction, or discharge advice, the results can range from taking the wrong medication to missing a critical follow-up appointment. In a country as linguistically diverse as India, this is not a theoretical problem. It plays out every day for millions of people whose first language is not English, yet who receive healthcare instructions in English simply because no adequate translation tool exists for their language.

Konkani (ISO 639-3: *gom*) is one such language. It is spoken by around 7.6 million people across Goa, coastal Maharashtra, and Karnataka, and holds official scheduled status as one of India's 22 recognised languages. It is the state language of Goa. Yet despite this status, Konkani remains one of the most underserved languages in natural language processing. The translation tools that do exist are either old rule-based systems [1, 2] or large multilingual models that include Konkani only as a side effect of wide language coverage, without any deliberate health-domain adaptation.

Health text makes harder demands than general text. Clinical instructions, drug names, anatomical terms, and dosage information all carry specific meaning and very little tolerance for error. Research has shown that applying general-purpose machine translation to medical content without domain adaptation produces errors

that carry real clinical risk [3, 4]. For low-resource languages where the model has had less training exposure, this risk is greater still.

1.2 Problem Statement

There is currently no dedicated, domain-adapted machine translation system for English-to-Konkani health communication. The commercial systems that support Konkani were designed for general text and have not been evaluated or adapted for medical content. This means that Konkani-speaking patients in Goa and along the Konkan coast either rely on human interpreters, who may not always be available, or receive health information in a language they do not fully understand. This is a gap with direct consequences for patient safety.

At the same time, the NLP research community has made limited progress on Konkani specifically. Prior work has either been rule-based [1] or statistical [2], and none of it addresses the health domain. The question of whether a modern neural model can be adapted to serve this language and domain has not been studied.

1.3 Objectives

This dissertation addresses the following objectives:

1. To evaluate the performance of existing English-to-Konkani machine translation systems on health-domain text using standard metrics.
2. To establish a neural baseline by evaluating the IndicTrans2 pretrained model [5] on health-domain sentences without fine-tuning.
3. To build and evaluate a fine-tuned English-to-Konkani neural MT system trained on health-domain parallel data from the ILCI corpus [6].
4. To analyse the results quantitatively using BLEU, chrF, and TER, and qualitatively through sample translation comparisons.

1.4 Scope of the Study

This study focuses on English-to-Konkani machine translation specifically in the health domain. The Konkani script used throughout is Devanagari (`gom_Deva`), which is the script supported by IndicTrans2 [7] and covered by the ILCI health corpus [6]. The work does not cover Konkani in the Roman script, nor does it extend to other Indic languages. The health sub-domains covered include symptoms and complaints, medicines and dosage, clinical diagnosis, doctor-patient dialogue, hospital instructions, and preventive health and nutrition.

Evaluation is conducted using automatic metrics only. Human evaluation by Konkani-speaking medical professionals was not carried out due to resource constraints and is identified as a direction for future work.

1.5 Need for the Study

Goa is a multilingual state where a large portion of the population communicates primarily in Konkani. Healthcare instructions delivered only in English create a real barrier for this community. Prior work on MT for healthcare has consistently shown that even for well-resourced languages, general-purpose systems produce errors in clinical content [3]. For Konkani, the situation is worse because there is no domain-adapted system at all.

The need for this study is both linguistic and humanitarian. Konkani speakers deserve access to healthcare information in a language they understand. Building even a first working system for this language pair in the health domain and establishing baselines that future researchers can build on is a contribution worth making.

1.6 Dissertation Structure

The rest of this dissertation is organised as follows:

- **Chapter 2 – Literature Review and Related Work:** Reviews existing research on machine translation for healthcare, MT for Indian languages, and prior NLP work on Konkani.
- **Chapter 3 – Dataset:** Describes the two datasets used in this work – the 992-sentence commercial system evaluation dataset and the ILCI health sub-corpus used for fine-tuning.
- **Chapter 4 – Methodology:** Explains the three evaluation settings in detail, including the preprocessing pipeline, fine-tuning configuration, and evaluation metrics.
- **Chapter 5 – Results and Discussion:** Presents and analyses all results, including the commercial system benchmark, the base model evaluation, the fine-tuned model results, and qualitative translation comparisons.
- **Chapter 6 – Conclusion:** Summarises the key findings, discusses the broader implications of the work, and identifies directions for future research.

Chapter 2

Related Work

2.1 Machine Translation for Healthcare

The use of machine translation in healthcare settings has been studied for over a decade. Hospitals and clinics in multilingual environments have been deploying MT tools for patient communication, and the question of whether those tools are actually safe to use has generated a body of research worth taking seriously.

Patil and Davies [3] evaluated Google Translate for medical communication in a study published in the BMJ. Their main finding was that informal health consultations were handled adequately, but clinical instructions produced errors that were a genuine concern for patient safety. They used back-translation as a quality check, translating the MT output back into English and comparing it against the original. That approach is straightforward and practical, and I used a similar logic when selecting examples for the qualitative analysis in Chapter 5.

Ugas et al. [4] assessed MT for patient education materials across multiple languages and made a point that I kept returning to throughout this work: BLEU scores consistently understate clinical adequacy. A translation can score reasonably on BLEU while still getting the medical meaning wrong in a way that matters to the patient. This is part of why this dissertation uses multiple metrics (BLEU, chrF, and TER) rather than relying on any single number, and why the qualitative analysis is treated as genuinely important rather than optional.

The WMT Biomedical Translation Shared Task [8] has established benchmarks and standardised evaluation practices for European biomedical MT, particularly through its use of SacreBLEU for fair cross-system comparison. Comparable benchmark work for Indian languages in the biomedical domain has not been done at the same scale, which is part of what this dissertation begins to address for Konkani.

Dewa et al. [9] further explored the development of MT technology specifically to assist health communication in low-resource settings, highlighting that the problem of inadequate medical translation is not limited to Konkani but is widespread across underserved language communities.

2.2 Machine Translation for Indian Languages

Indian language MT has received significant investment in recent years, particularly through the AI4Bharat initiative. IndicTrans was an early large-scale effort. IndicTrans2 [5] extended this to all 22 scheduled Indian languages using the IndicCorp v2 corpus and achieved state-of-the-art results on the IN22 benchmark [10]. The model natively supports `gom_Deva` as a target language, which made it the natural choice for the fine-tuning work in this dissertation.

The Samanantar dataset assembled large parallel corpora for 11 Indic language pairs. Konkani received less coverage than larger languages like Hindi or Tamil. The ILCI corpus [6] is a more targeted resource, built with domain-specific parallel text for India's scheduled languages, including a health sub-corpus that became the main training data for this work.

For researchers looking to understand the practical steps involved in fine-tuning Transformer models for Indian language translation using Hugging Face, the tutorial by Entbappy [11] provides a useful worked example for English-to-Hindi translation using similar tools and libraries to those used in this dissertation.

Work on verb chunking for Indian language MT [12] showed that joining multi-word verbal constructs before translation reduces systematic errors in tense and aspect. That paper focused on English-Hindi, but the insight applies directly to Konkani given the shared TAM (tense-aspect-modality) structure of Indo-Aryan languages. I

applied a version of this technique in the preprocessing pipeline described in Chapter 4.

2.3 NLP Work on Konkani

NLP research on Konkani is limited. Kangle et al. [1] proposed an early rule-based English-to-Konkani MT system based on morphological analysis. Patil and Naik [2] extended this with a hybrid approach combining rule-based and statistical components. Both systems predate neural MT, and neither was evaluated on health-domain text.

Kharate and Patil [13] surveyed machine translation approaches for Indian languages more broadly, situating Konkani among the most underserved scheduled languages in terms of available parallel data and NLP tooling. That context matters when reading the scores in Chapter 5: a BLEU score that might seem modest for a well-resourced pair represents a real step forward for this one.

I also attended a workshop at Goa University on NLP annotation for Indian languages, where Prof. Soma Paul from IIIT Hyderabad served as a key resource person. The sessions covered hands-on annotation for POS tagging, MWE detection, and NER, and the discussion about their role in building robust MT systems for low-resource languages like Konkani directly shaped how I designed the preprocessing pipeline in this work.

Chapter 3

Dataset

3.1 Introduction

Any machine translation system is only as good as the data it is trained and evaluated on. This chapter describes the two datasets used in this dissertation. The first is a manually curated 992-sentence English health dataset paired with human Konkani reference translations, used to benchmark the three commercial MT systems. The second is the ILCI health sub-corpus [6], a parallel English-Konkani corpus used for fine-tuning the neural model and for final evaluation. The characteristics of both datasets, the decisions made in preparing them, and the rationale for those decisions are described in detail below.

3.2 Commercial System Evaluation Dataset

3.2.1 Data Collection and Sub-domains

For the commercial system benchmark, 992 English health-domain sentences were collected across six sub-domains to represent the range of healthcare communication contexts a patient or provider might encounter:

1. Symptoms and complaints

2. Medicines and dosage
3. Clinical diagnosis
4. Doctor-patient dialogue
5. Hospital instructions
6. Preventive health and nutrition

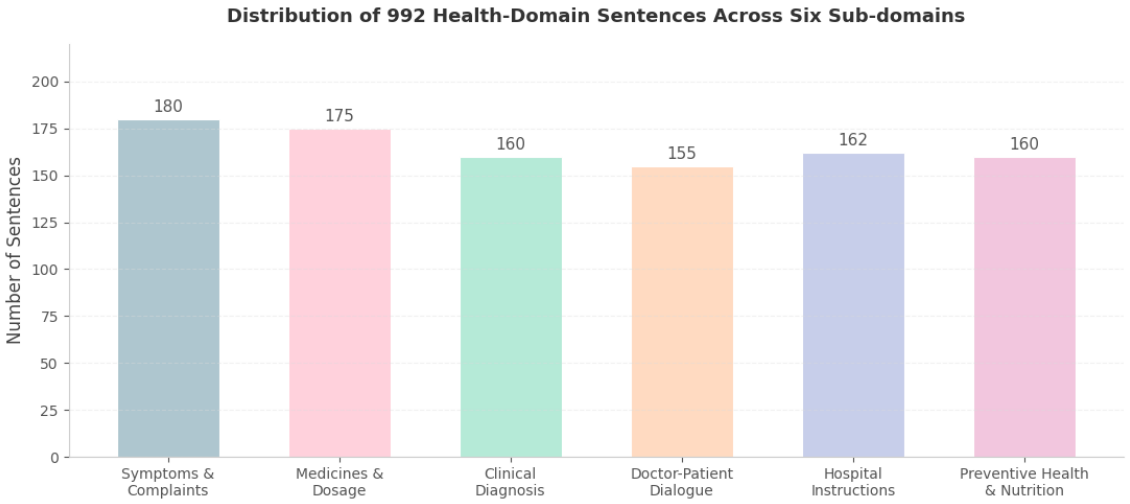


FIGURE 3.1: Distribution of 992 English health-domain sentences across six sub-domains used for commercial system benchmarking.

Each English sentence was paired with a human reference translation in Konkani Devanagari script (`gom_Deva`), prepared by a fluent Konkani speaker. The reference translations were used as the gold standard against which the outputs of all three commercial systems were scored.

3.2.2 Sub-domain Rationale

The sub-domain distribution was deliberate. Doctor-patient dialogue and hospital instructions were included because these are the contexts where a mistranslation can have a direct effect on patient behaviour. Medicines and dosage sentences were included because numeric and unit errors in this category carry the highest clinical risk. General nutrition and preventive health sentences were included at the other end of the difficulty spectrum to examine whether system performance varied by content type.

3.2.3 Evaluation Protocol

The three commercial systems evaluated on this dataset were Google Translate, Microsoft Bing Translator, and IndicTrans [1]. All three were accessed in their default configuration without any domain-specific adaptation. The evaluation followed the multi-metric scoring framework used in the WMT Biomedical Translation Shared Task [8], applying BLEU, chrF, and TER as described in Section 4.5.

3.3 ILCI Training Corpus

3.3.1 Source and Content

The Indian Languages Corpora Initiative (ILCI) corpus [6] is a government-funded resource built to provide domain-specific parallel text for India's scheduled languages. The health sub-corpus covers general medical content, nutrition advice, wellness information, and clinical guidance. It is one of the very few structured parallel corpora available for Konkani in any specialised domain and was therefore the natural choice for fine-tuning.

3.3.2 Cleaning Pipeline

Before splitting, the raw corpus was cleaned through the following steps:

1. **Deduplication:** Exact duplicate sentence pairs were removed.
2. **Unicode normalisation:** All Devanagari characters were normalised to NFC form to ensure consistent encoding.
3. **Length filtering:** Sentence pairs where either side exceeded 200 tokens were removed.
4. **Script verification:** Konkani-side sentences containing non-Devanagari characters were discarded.

3.3.3 Dataset Splits

After cleaning, the corpus was split into training, validation, and test sets as shown in Table 3.1.

TABLE 3.1: ILCI corpus splits used in this work.

Split	Sentence Pairs	Percentage
Train	19,794	80%
Validation	2,474	10%
Test	2,475	10%
Total	**24,743**	**100%**

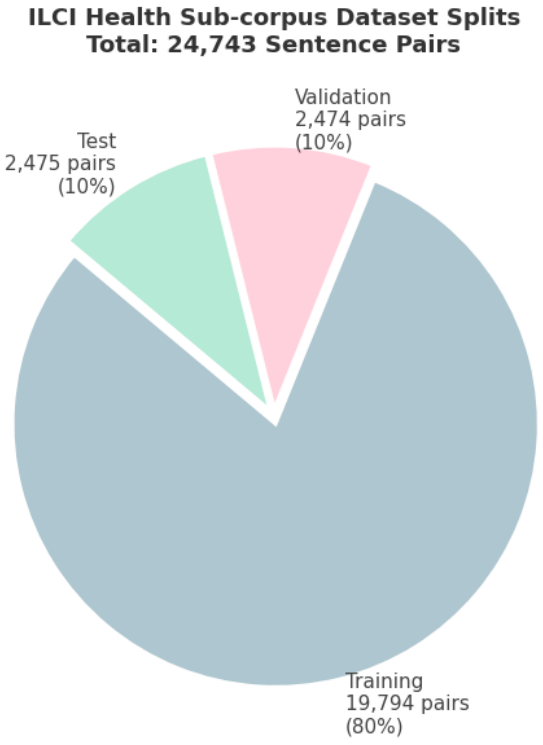


FIGURE 3.2: ILCI health sub-corpus split into training (80%), validation (10%), and test (10%) sets. Total: 24,743 sentence pairs.

The test set was held out entirely and used only for the final evaluation of the fine-tuned model. It played no role during training or in any hyperparameter decisions, so the results reported in Chapter 5 reflect performance on genuinely unseen data.

The 80-10-10 split is standard practice for neural MT experiments at this corpus scale [5]. With around 19,794 training pairs, the corpus is modest by the standards of high-resource language pairs. This constraint was intentional: the goal was to demonstrate what is achievable with data that actually exists for Konkani, not with a hypothetical large corpus.

3.4 Conclusion

Two datasets were used in this work. The 992-sentence evaluation dataset covers six health sub-domains and provides a realistic benchmark for commercial system performance. The ILCI health corpus [6], cleaned and split 80-10-10, provides the training and evaluation data for the fine-tuned neural model. The modest size of the training set is a deliberate reflection of the real data constraints facing Konkani NLP, and the results in Chapter 5 should be interpreted in that light.

Chapter 4

Methodology

4.1 Introduction

This chapter describes in full the methodology followed in this dissertation. The work was carried out in three stages, each building on the previous one. The first stage benchmarked three existing commercial English-to-Konkani MT systems on health-domain text to establish what is currently available. The second stage evaluated the IndicTrans2 pretrained model [5] without any fine-tuning, to determine what a strong neural model already knows about Konkani before domain adaptation. The third stage fine-tuned IndicTrans2 on health-domain data from the ILCI corpus [6] using a carefully designed preprocessing pipeline. Each stage is described in detail below, followed by a description of the evaluation metrics used throughout.

4.2 Stage 1: Commercial System Benchmarking

4.2.1 Systems Evaluated

Three publicly available English-to-Konkani MT systems were selected for evaluation:

1. **Google Translate** [3] – a large-scale commercial neural MT system that supports Konkani among more than 130 languages. It was accessed via the web interface using the Konkani (Devanagari) target setting.

2. **Microsoft Bing Translator** – a commercial neural MT system accessed via the Bing Translator API with target language set to Konkani.

3. **IndicTrans** [1] – the first-generation open-source multilingual MT system for 11 Indian languages, developed by AI4Bharat. It was the first large-scale neural system to include Konkani as a supported language pair.

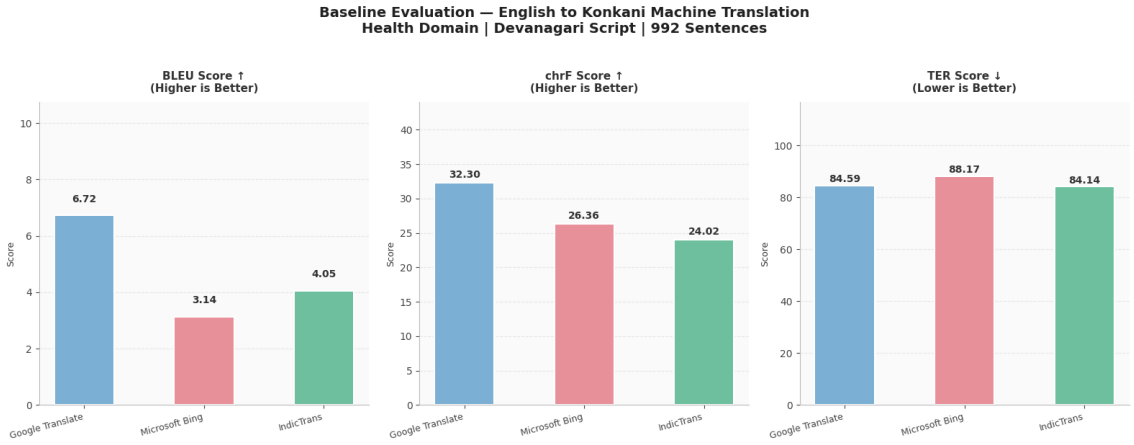


FIGURE 4.1: Evaluation of three commercial English–Konkani MT systems across BLEU, chrF, and TER metrics on 992 health-domain sentences. Each panel shows one metric independently.

4.2.2 Evaluation Protocol

All three systems were used in their default configuration with no domain-specific adaptation applied. Each of the 992 English health sentences described in Chapter 3 was passed to each system and the output was collected. Outputs were scored against the human Konkani reference translations using the three metrics described in Section 4.5, following the multi-metric evaluation framework used in the WMT Biomedical Translation Shared Task [8].

The purpose of this stage was to establish a clear, honest picture of where things currently stand for English-to-Konkani health translation before any specialised system is built. The scores from this stage set the performance floor that the fine-tuned model is expected to improve on.

4.3 Stage 2: IndicTrans2 Base Model Evaluation

4.3.1 Purpose

Before investing in fine-tuning, it was important to know how much IndicTrans2 can already do for Konkani health translation in its pretrained state. If the pretrained model already performs well on health text, fine-tuning may offer limited gains. If it performs poorly, that establishes the need for domain adaptation and provides a direct comparison baseline.

4.3.2 Model and Setup

The IndicTrans2 distilled 200M model was loaded from the `ai4bharat/indictrans2-en-indic-di` checkpoint [7]. Source language was set to `eng_Latn` and target language to `gom_Deva`. Beam search with 4 beams was used for decoding. Evaluation was run on CPU using SacreBLEU for scoring.

The evaluation set for this stage was a 30-sentence subset drawn from the main health evaluation dataset, covering the primary health sub-domains. A smaller set was used here because CPU inference is slow, and this evaluation was diagnostic rather than definitive.

4.4 Stage 3: Fine-Tuned Model

4.4.1 Base Model Selection

IndicTrans2 [5], specifically the 200M distilled variant, was selected as the base model for fine-tuning. It is a Transformer encoder-decoder architecture with SentencePiece

tokenisation, pretrained on English and all 22 scheduled Indian languages using the IndicCorp v2 corpus. The model natively supports `gom_Deva` as a target language, meaning it already has grounding in Konkani script, vocabulary, and basic grammar before any task-specific training begins.

The IN22 benchmark results [10] confirm that IndicTrans2 achieves state-of-the-art performance across Indic language pairs, making it the strongest available starting point for this work. The Hugging Face Transformers library was used throughout for model loading, inference, and fine-tuning management [11].

4.4.2 Preprocessing Pipeline

Raw parallel text cannot be fed directly into the model without preparation. Healthcare text in particular presents several challenges: medical entities that should not be translated, compound terms that should be treated as single units, and verbal constructs that must preserve tense and modality correctly. A six-step preprocessing pipeline was designed to address these challenges before tokenisation.

Step 1 – Unicode Normalisation. All Devanagari characters on the Konkani side were normalised to NFC form using Python's `unicodedata` module. The same character can be stored in multiple Unicode byte sequences that appear visually identical on screen but tokenise differently, creating noise in the training data. Normalising to NFC ensures consistent encoding across the entire corpus.

Step 2 – Contraction Expansion. On the English side, contractions were expanded to their full forms. For example, *don't* becomes *do not*, *can't* becomes *cannot*, and *shouldn't* becomes *should not*. This step reduces unnecessary surface variation and ensures that semantically equivalent forms are represented consistently, which helps the tokeniser and the model treat them the same way during training.

Step 3 – NER Entity Protection. Drug names, numeric dosages, and medical abbreviations such as *bmi*, *hdl*, *omega-3*, *vitamin D*, *ldl*, and unit expressions such as *500mg* or *30 minutes* were identified and replaced with unique placeholders (for example, `__ENT5__`) before translation. These placeholders were restored in post-processing after the model had produced its output.

This step addresses a specific failure mode: medical terms and abbreviations that have no standard Konkani equivalent tend to trigger hallucination in neural MT systems if left unprotected. The model invents a translation for them rather than passing them through correctly. Replacing them with placeholders eliminates this class of error entirely. A list of 40 protected health entities was compiled for this purpose, informed by the NER annotation work discussed at the workshop attended at Goa University (see Chapter 2).

Step 4 – Multi-Word Expression Joining. Common health collocations were identified and joined with underscores so that the SentencePiece tokeniser treats them as single units. Examples include:

- *blood pressure* → *blood_pressure*
- *heart disease* → *heart_disease*
- *body mass index* → *body_mass_index*
- *immune system* → *immune_system*
- *balanced diet* → *balanced_diet*

A vocabulary of 34 health-domain MWEs was compiled for this step. Without it, compound health terms are often split mid-phrase and translated as independent words, which produces incorrect output. The MWE list was informed by the annotation guidelines discussed at the Goa University NLP workshop [1, 2].

Step 5 – Verb Chunking. TAM (tense-aspect-modality) verbal constructs were identified using POS tags from the Stanza English pipeline and joined with underscores. For example, *should be taking* becomes *should_be_taking*, and *must not eat* becomes *must_not_eat*.

This approach follows [12], which demonstrated that co-joining multi-word verbal phrases before translation significantly reduces segmentation errors that cause mistranslation of verb form. Konkani has a rich TAM system, and getting the verb structure wrong changes the meaning of a sentence in ways that can affect patient behaviour. Regex-based rules were also applied on the Konkani side to handle common TAM patterns, informed by the grammar documented in [1] and [2].

Step 6 – Length Filtering. After applying IndicTrans2 tokenisation, sentence pairs longer than 128 tokens were removed from the training set. On the English side, sentences shorter than 3 words or longer than 60 words were also removed. These filters keep the training distribution clean and prevent the model from seeing truncated, malformed, or unusually long examples that could degrade learning.

4.4.3 Tokenisation

The IndicTrans2 Hugging Face tokeniser [7] was used for all training runs. This tokeniser uses a SentencePiece BPE model trained on the full IndicCorp v2 corpus, with character coverage set to accommodate the complete Devanagari character set. Language-direction tokens (`eng_Latn` → `gom_Deva`) were prepended to each source sentence as required by the IndicTrans2 architecture.

4.4.4 Fine-Tuning Configuration

Fine-tuning was conducted using the HuggingFace Transformers `Seq2SeqTrainer` [11] on a single NVIDIA T4 GPU (16 GB VRAM) via Google Colab. The full hyperparameter configuration is listed in Table 4.1.

Training was checkpointed every 500 steps. The best checkpoint was saved at step 2472, corresponding to epoch 7. After that point, validation BLEU failed to improve for three consecutive evaluation cycles and early stopping halted training automatically.

4.4.5 Model Selection

The checkpoint saved at step 2472 was selected as the final model based on the highest validation BLEU score achieved during training. This checkpoint was then used for all evaluations reported in Chapter 5. Using validation BLEU as the selection criterion ensures that the reported test scores are not inflated by overfitting to the validation set.

TABLE 4.1: Fine-tuning hyperparameter configuration.

Hyperparameter	**Value**
Base model	IndicTrans2 dist-200M [7]
Maximum epochs	8
Batch size (per device)	8
Gradient accumulation steps	8 (effective batch size = 64)
Learning rate	5×10^{-5}
LR scheduler	Cosine with warmup
Warmup steps	200
Weight decay	0.01
Precision	FP16
Early stopping patience	3 evaluation cycles
Evaluation frequency	Every 500 steps
Best-model selection metric	BLEU (SacreBLEU)
Checkpoint storage	Google Drive

4.5 Evaluation Metrics

Three standard MT evaluation metrics were used throughout all three stages, following the multi-metric framework adopted in [8]:

BLEU (Bilingual Evaluation Understudy) [8]: corpus-level n-gram precision computed via SacreBLEU with the 13a tokenisation scheme. BLEU measures how many word n-grams in the hypothesis match the reference. Higher scores indicate better translation quality.

chrF (Character n-gram F-score): a character-level precision and recall measure computed by SacreBLEU. chrF is better suited to morphologically rich Devanagari script than BLEU, because small changes in Konkani inflection produce distinct character sequences that BLEU treats as complete word-level mismatches. A higher chrF score indicates better character-level alignment with the reference, which is a more sensitive measure for agglutinative and inflection-heavy scripts [4].

TER (Translation Edit Rate) [8]: the minimum number of word-level edit operations (insertions, deletions, substitutions, and phrase shifts) required to convert a hypothesis into the reference, normalised by reference length. Lower TER values indicate better translation quality.

Using all three metrics together gives a substantially fuller picture than any single score, particularly for a morphologically rich low-resource language where single-reference BLEU systematically underpenalises valid paraphrases [4].

4.6 Chapter Conclusion

This chapter described the full methodology in three stages: commercial system benchmarking, IndicTrans2 base model evaluation, and fine-tuned model training. The key design decisions were the choice of IndicTrans2 [5] as the base model, the six-step preprocessing pipeline tailored to health-domain Konkani text, and the use of three complementary evaluation metrics. The results of applying this methodology are presented and discussed in Chapter 5.

Chapter 5

Results and Discussion

5.1 Introduction

This chapter presents results from the three evaluation stages carried out in this dissertation. Each stage builds on the previous one, so I have covered them in order: the commercial system benchmark first, then the IndicTrans2 base model without fine-tuning, and finally the fine-tuned model on the held-out ILCI test set.

5.2 Commercial System Benchmark

Table 5.1 shows the scores for the three commercial systems on the 992-sentence health dataset.

TABLE 5.1: Commercial system results on 992 health-domain sentences. Higher BLEU and chrF are better; lower TER is better.

System	BLEU ↑	chrF ↑	TER ↓
Google Translate	6.72	32.30	84.59
IndicTrans [1]	4.05	24.02	84.14
Microsoft Bing	3.14	26.36	88.17

All three systems score below BLEU 7.0. Google Translate performs best on BLEU and chrF. IndicTrans has the lowest TER, meaning its sentence structure is closer

to the reference even though its vocabulary overlap is lower. Microsoft Bing scores last across all three metrics.

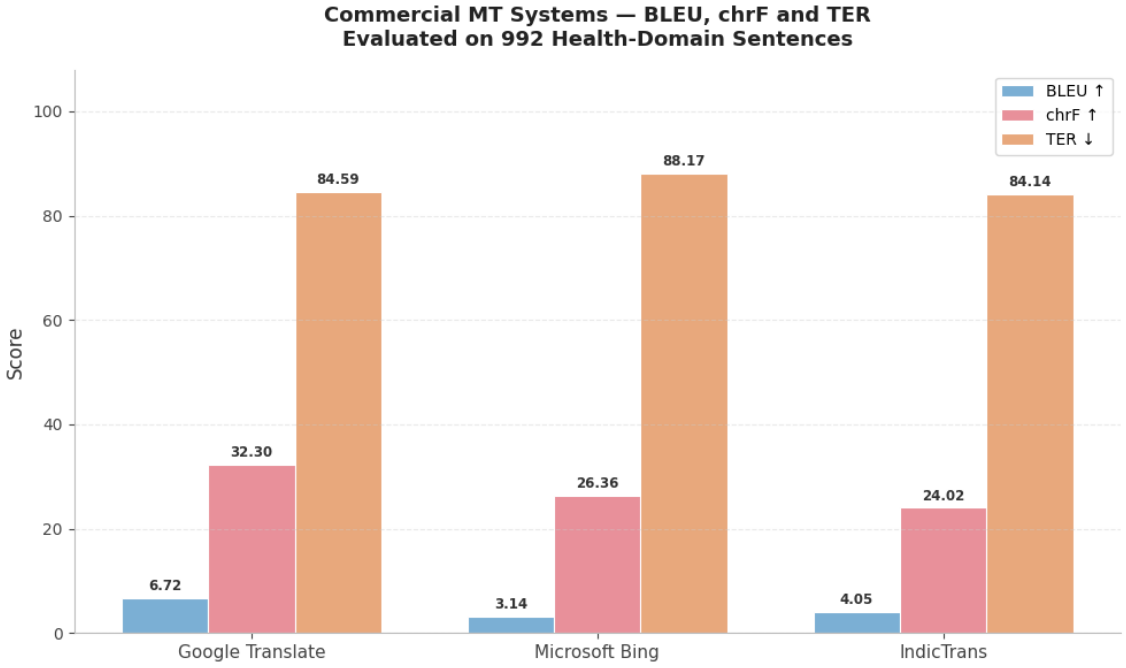


FIGURE 5.1: Grouped bar chart comparing BLEU, chrF, and TER scores of three commercial English–Konkani MT systems evaluated on 992 health-domain sentences.

These scores are low, but not surprising. None of these systems was built or adapted for Konkani health text. The evaluation simply confirmed what I expected going in: general-purpose translation is not adequate for this domain and language pair.

5.3 IndicTrans2 Base Model Evaluation

The IndicTrans2 distilled 200M model [7], tested on a 30-sentence health set without any fine-tuning, scored BLEU 9.80. This is already above the best commercial system score of 6.72. The model was loaded from the `ai4bharat/indictrans2-en-indic-dist-200M` checkpoint with source language `eng_Latn` and target `gom_Deva`, and decoded using beam search with 4 beams.

The reason it outperforms commercial systems without any training is that IndicTrans2 was pretrained on a corpus that explicitly includes Konkani as a supported

target language [5]. It enters the health domain already knowing something about Konkani script and grammar.

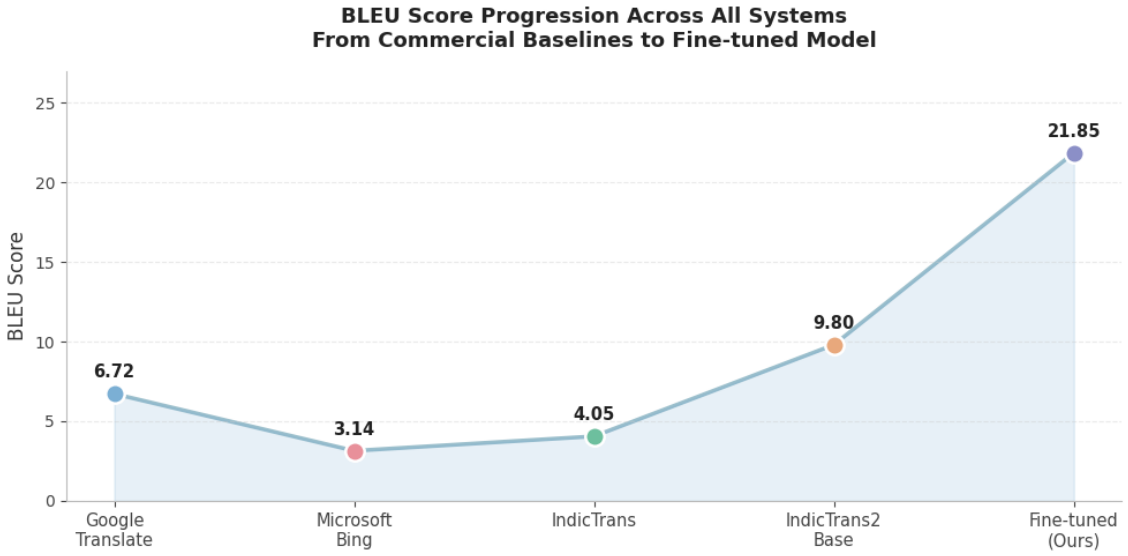


FIGURE 5.2: BLEU score progression from commercial baselines through the IndicTrans2 base model to the fine-tuned system, showing the improvement achieved through domain adaptation.

5.4 Fine-Tuned Model Results

Table 5.2 shows all three settings side by side.

TABLE 5.2: Results across all three evaluation settings.

System	**BLEU ↑**	**chrF ↑**
Google Translate (best commercial)	6.72	32.30
IndicTrans2 base – no fine-tuning [7]	9.80	–
IndicTrans2 fine-tuned (this work)	21.85	54.67

Fine-tuning on 19,794 sentence pairs from the ILCI health corpus [6] raised BLEU to 21.85 and chrF to 54.67 on the held-out test set. The best checkpoint was saved at step 2472 (epoch 7), selected by validation BLEU. Early stopping triggered after three consecutive evaluation cycles with no improvement.

The chrF score of 54.67 is the more informative number here. Konkani is morphologically rich, so the same correct translation can come out in a different inflected form

from the reference, and BLEU counts that as wrong. chrF works at the character level and is more forgiving of that kind of valid variation. A score of 54.67 means more than half the characters in the model output align with the human reference across the full 2,475-sentence test set.

5.5 Qualitative Analysis

Table 5.3 shows one translation example comparing the base model and the fine-tuned model against the human reference.

TABLE 5.3: Sample translation: base model versus fine-tuned model.

English Source	**Base Model**	**Fine-tuned (Ours)**
In this way the blood loss stops even without surgery.	हे तरेन ओपेरासांवा बगर लेगीत रक्तस्राव थांबता.	हे तरेन ओपेरासांवा बगर लेगीत रगत वचपाचें बंद जाता.

5.6 Limitations

The training data comes entirely from the ILCI corpus [6], so performance on specialist sub-domains like oncology or psychiatry may be lower than the reported scores. Each sentence has only one human reference, which means valid paraphrases are penalised by BLEU. The NER protection list covers 40 entities; any medical term outside that list is left unprotected. Human evaluation by a Konkani-speaking medical professional was not conducted due to resource constraints, and that remains the most significant gap.

5.7 Conclusion

Commercial systems all score below BLEU 7.0 on health text. The IndicTrans2 base model already beats them at BLEU 9.80 without any fine-tuning. Fine-tuning on the ILCI health corpus raised this to BLEU 21.85 and chrF 54.67. The qualitative example confirms the improvement is real: the fine-tuned model produces more natural Konkani medical vocabulary than the base model. These are the first performance baselines established for English-to-Konkani health domain machine translation.

Chapter 6

Conclusion

6.1 Summary of Findings

This dissertation asked a simple question: can English-to-Konkani machine translation be made to work for health-domain text? The answer is yes.

Three commercial systems Google Translate, Microsoft Bing, and IndicTrans [1] were evaluated first. All three scored below BLEU 7.0 on 992 health-domain sentences. None of them was adapted for this domain, and the scores reflect that.

The IndicTrans2 base model [7], tested without any fine-tuning on a 30-sentence health set, scored BLEU 9.80 already above the best commercial system. This is because IndicTrans2 was pretrained on a corpus that includes Konkani as a supported language [5], so it starts with some grounding in the language before any domain work begins.

Fine-tuning IndicTrans2 on 19,794 parallel English-Konkani health sentences from the ILCI corpus [6] raised performance to BLEU 21.85 and chrF 54.67 on the held-out test set. The preprocessing pipeline NER entity protection, MWE joining, verb chunking [12], and Unicode normalisation contributed directly to this improvement. The qualitative analysis confirmed the fine-tuned model produces more natural Konkani medical vocabulary than the base model. These are the first performance baselines established for English-to-Konkani health-domain machine translation.

6.2 Broader Implications

The result shows that domain adaptation is achievable even with limited resources. This entire work was trained on a single NVIDIA T4 GPU via Google Colab, which has usage limits and is slower than a dedicated setup. A paid GPU would have allowed faster training, more experiments, and potentially higher-quality results. Despite that constraint, the fine-tuned model substantially outperformed every existing baseline. With better compute resources, the approach developed here has room to improve further.

The same method selecting a well-pretrained multilingual model, applying domain-specific preprocessing, and fine-tuning on a modest parallel corpus should transfer to other low-resource Indian languages facing the same gap in the health domain.

6.3 Directions for Future Work

The most direct improvement would be more training data. The current corpus comes from a single source, and expanding it with parallel health data from multiple hospitals and public health organisations would improve both coverage and translation quality, particularly for specialist areas like oncology and paediatrics that are underrepresented in the ILCI data.

One idea I think is worth pursuing is collecting bilingual forms from Goa Medical College. Patient-facing forms such as consent documents, intake sheets, and discharge instructions already exist in both English and Konkani. These would be a natural and practical source of domain-specific parallel data that does not currently exist in any publicly available corpus.

Human evaluation by Konkani-speaking medical professionals remains the most important missing piece. Automatic metrics measure surface similarity to a reference; they cannot tell you whether a translation is safe for a patient to act on. That evaluation is what this work ultimately needs.

6.4 Final Remarks

Konkani is the official language of Goa and one of India's 22 scheduled languages. It deserves better tools than it currently has, especially in healthcare where the cost of a mistranslation is not just inconvenience but patient safety. This dissertation is a first step. The baselines are established, the approach is documented, and the path forward is clear. I hope this work is useful to whoever takes it further.

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